

Xiaoyang Zhan

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA

Ph.D., Mechanical Engineering (Robotics) — CERLAB, Advisor: Prof. Kenji Shimada

In Progress

M.S., Mechanical Engineering — GPA: 3.96/4.0

2023

University of Pittsburgh — Pittsburgh, PA (dual-degree with Sichuan University)

B.E., Mechanical Engineering, Minor in Computer Science — GPA: 3.94/4.0

2021

RESEARCH PROFILE

CMU Ph.D. researcher focused on decision-making and planning for autonomous robots — from single-robot active perception to heterogeneous multi-robot collaboration between aerial and legged platforms. Broad supporting experience in perception and spatial representation (semantic mapping, LiDAR-camera fusion, real-time dynamic-object tracking), with hands-on integration and real-world validation on ground and aerial platforms. Current work connects embodied navigation and spatial understanding through pose-aware sensing and VLM-assisted decision-making, directly applicable to embodied AI and autonomous-driving planning stacks. Mentors a team of up to eight.

SELECTED PUBLICATIONS

- **X. Zhan***, S. Chen*, K. Shimada. "Pose-Aware Legged Robot Semantic Exploration with Omnidirectional Perception in Confined Unknown Environments." *arXiv preprint*, 2026. (*co-first author; submitted to ICRA 2027)
- **X. Zhan**, S. Zhou, Q. Yang, Y. Zhao, H. Liu, S. C. Ramineni, K. Shimada. "Semantic Exploration and Dense Mapping of Complex Environments Using Ground Robot with Panoramic LiDAR-Camera Fusion." *IEEE Robotics and Automation Letters (RA-L)*, 2025. (1st author)
- Z. Xu*, **X. Zhan***, Y. Xiu, C. Suzuki, K. Shimada. "Onboard Dynamic-Object Detection and Tracking for Autonomous Robot Navigation with RGB-D Camera." *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 1, pp. 651–658, Jan. 2024. (100+ citations; *co-first author)
- Z. Xu*, **X. Zhan***, B. Chen, Y. Xiu, C. Yang, K. Shimada. "A Real-Time Dynamic Obstacle Tracking and Mapping System for UAV Navigation and Collision Avoidance with an RGB-D Camera." *IEEE International Conference on Robotics and Automation (ICRA)*, 2023. (*co-first author)

RESEARCH EXPERIENCE

Ph.D. Researcher, CERLAB — Carnegie Mellon University, Advisor: Prof. Kenji Shimada

2022 – Present

Pose-Aware Active Perception and VLM-Assisted Decision-Making

submitted to ICRA 2027

Video: [YouTube](#)

- Conceived and led the research direction, designing a legged-robot system that jointly selects observation viewpoints and body posture based on target visibility and accumulated spatial observations.
- Guided the integration of a vision-language model with object-centric spatial memory and multi-view observation history to prune redundant viewpoints, translating multimodal judgments into executable planning decisions
- Improved target-surface coverage by 8–10 percentage points and cut exploration time by 17–32% versus a no-pose-adjustment baseline across three simulated scenes; validated on Boston Dynamics Spot in a real machine shop

Semantic Mapping and Hierarchical Planning for Autonomous Navigation

RA-L 2025

Code: [SEDEM](#)

- Led the research and implementation of an autonomous exploration system spanning LiDAR-camera fusion, dense semantic target mapping, viewpoint selection, and hierarchical planning; validated the integrated system in simulation and real-world unstructured environments
- Designed a planner that jointly schedules geometric coverage and multi-view target observations, using decoupled viewpoint sampling to satisfy inspection requirements while reducing redundant travel; outperformed evaluated baselines in exploration speed and path length
- Built a modular semantic mapping pipeline from panoramic imagery, LiDAR point clouds, and SLAM odometry to construct spatial representations for planning; added an execution state machine with adaptive sampling and recovery behavior for safe extension into unknown space

Heterogeneous Air-Ground Planning and Collaboration

manuscript in preparation — Sept. 2026

- Designing a UAV-Legged robot collaborative framework with platform-aware spatial representation and planning matched to each robot's sensing and mobility profile
- Developing a graph-based structure for sharing locally processed information across robots to reduce inter-robot communication bandwidth while preserving map consistency, evaluated across three real-world scenarios

Real-Time Dynamic Perception and Mapping for UAV Navigation

RA-L 2024 & ICRA 2023

Code: [onboard_detector](#) | [map_manager_pub](#)

- Co-first-authored a lightweight, real-time RGB-D dynamic-object detection and tracking pipeline for compute-constrained robots; achieved 0.11 m position error and 0.23 m/s velocity error onboard a small quadcopter, outperforming baselines, and released it as an open-source ROS package
- Co-first-authored a hybrid mapping system combining Kalman-filter state estimation, static occupancy mapping, and Markov-chain trajectory prediction, demonstrating safe UAV navigation in cluttered environments at 25 Hz on an NVIDIA Jetson Xavier

TECHNICAL SKILLS

- **Languages and Tools:** C++, Python, Shell, Git/GitHub, Linux (Ubuntu), Docker
- **Robotics & Autonomy:** ROS (multi-robot networking), MAVROS/PX4, Spot SDK, FAST-LIO2, occupancy mapping, A*/Dijkstra, LKH (TSP), CMU Autonomous Exploration stack, TARE
- **Simulation & Sim-to-Real:** NVIDIA Isaac Sim, Isaac Lab, Gazebo, Habitat
- **Perception & ML:** PyTorch, OpenCV, Segment Anything (SAM); Sensor fusion & calibration
- **Hardware Platforms:** Boston Dynamics Spot, custom UAV, Livox/Ouster LiDAR, Intel RealSense, fisheye camera (Insta360), NVIDIA Jetson